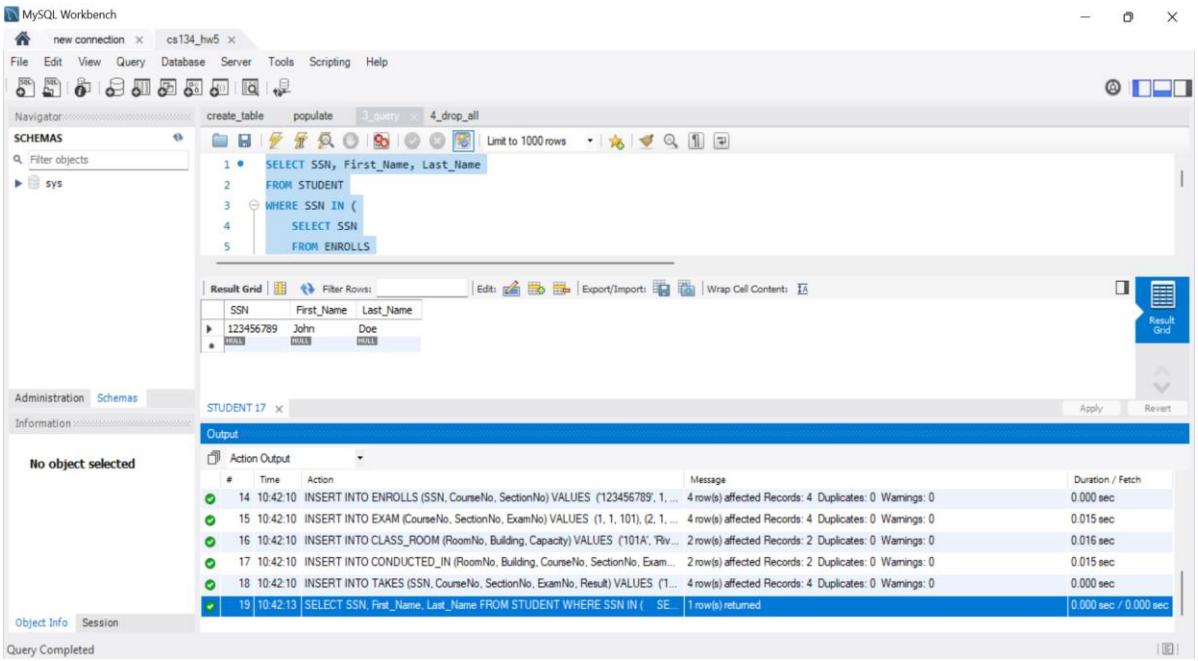
Kanageshwaran Dhakshinamoorthy
CSC 134
December 1, 2024
Homework 5:

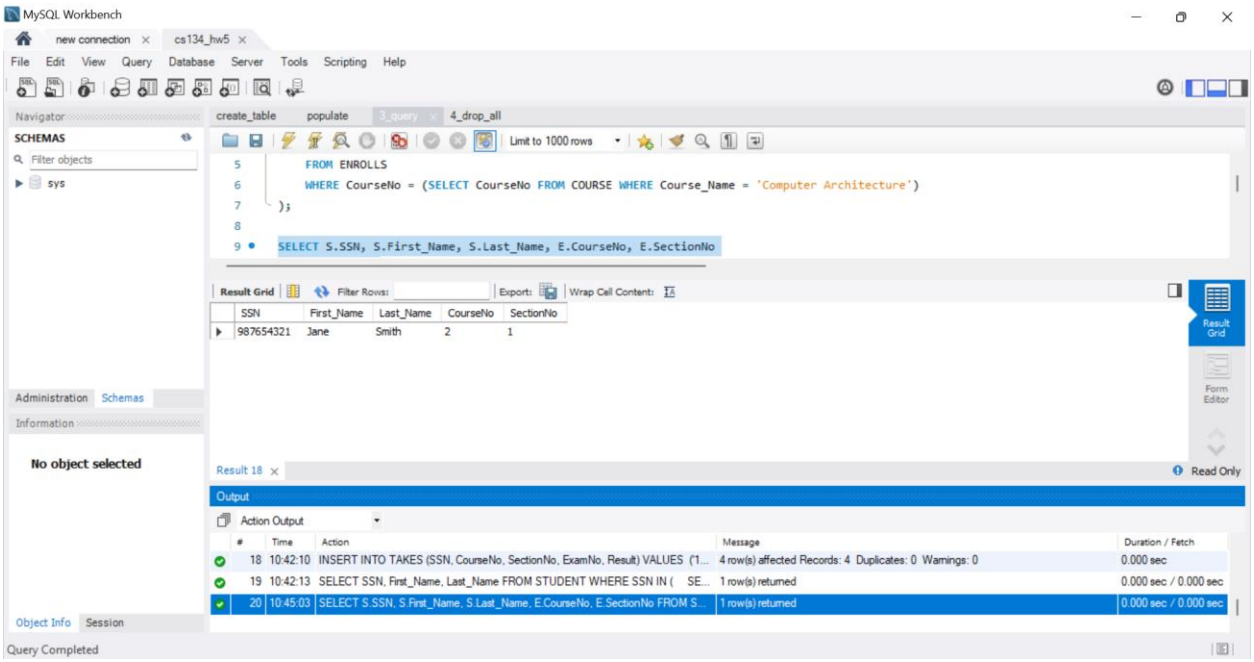
Query 1. Retrieve information (SSN, first name and last name) about students who took the course ‘Computer Architecture’ (“Computer Architecture” is a course name).

```
SELECT SSN, First_Name, Last_Name
FROM STUDENT
WHERE SSN IN (
  SELECT SSN
  FROM ENROLLS
  WHERE CourseNo = (SELECT CourseNo FROM COURSE WHERE Course_Name =
'Computer Architecture')
);
```



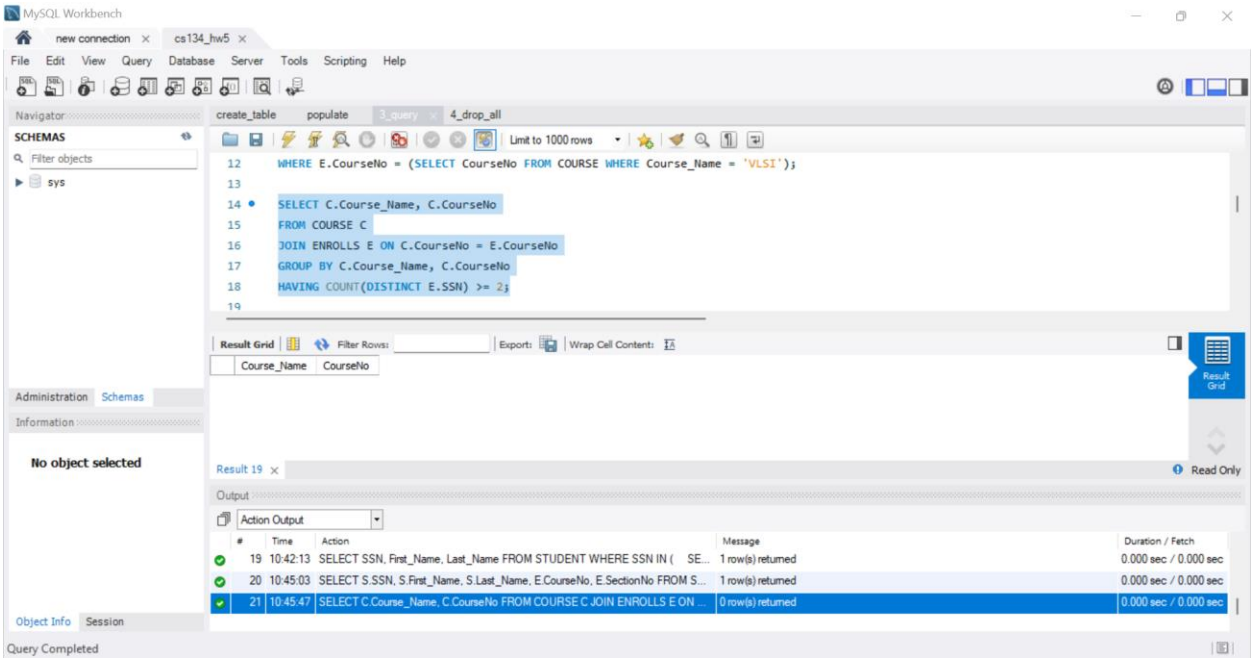
Query 2. Get the information (SSN, first name and last name) about students who have taken the course ‘VLSI’. Also get the course number and the section number in which they have enrolled in the course.

```
SELECT S.SSN, S.First_Name, S.Last_Name, E.CourseNo, E.SectionNo
FROM STUDENT S
JOIN ENROLLS E ON S.SSN = E.SSN
WHERE E.CourseNo = (SELECT CourseNo FROM COURSE WHERE Course_Name = 'VLSI');
```



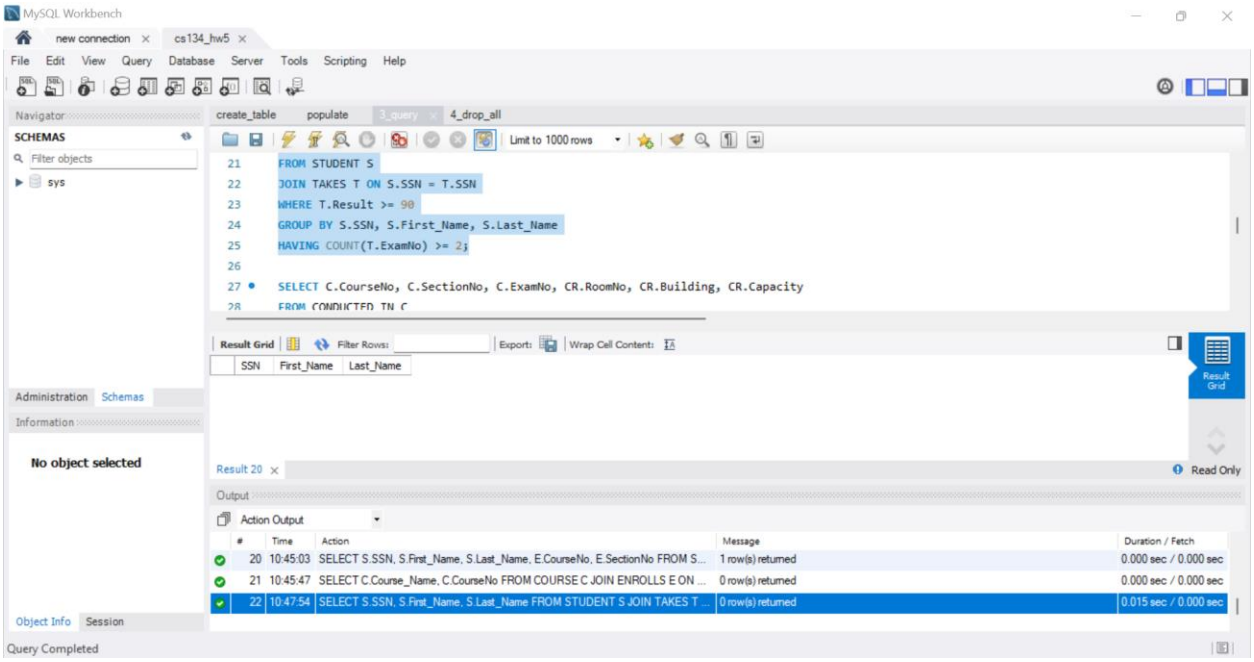
Query 3. Get the course names and course numbers which have 2 or more students.

```
SELECT C.Course_Name, C.CourseNo
FROM COURSE C
JOIN ENROLLS E ON C.CourseNo = E.CourseNo
GROUP BY C.Course_Name, C.CourseNo
HAVING COUNT(DISTINCT E.SSN) >= 2;
```



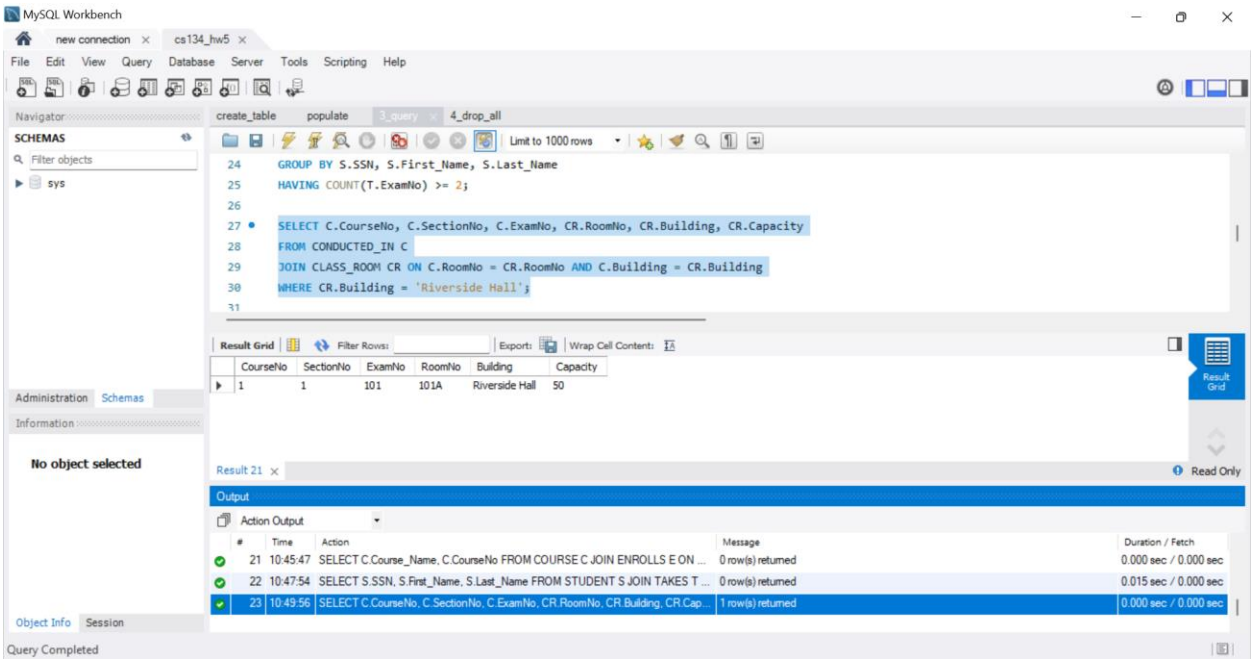
Query 4. Retrieve the information (SSN, first name and last name) of students who got Grade ‘A’ in at least two exams.

```
SELECT S.SSN, S.First_Name, S.Last_Name
FROM STUDENT S
JOIN TAKES T ON S.SSN = T.SSN
WHERE T.Result >= 90
GROUP BY S.SSN, S.First_Name, S.Last_Name
HAVING COUNT(T.ExamNo) >= 2;
```



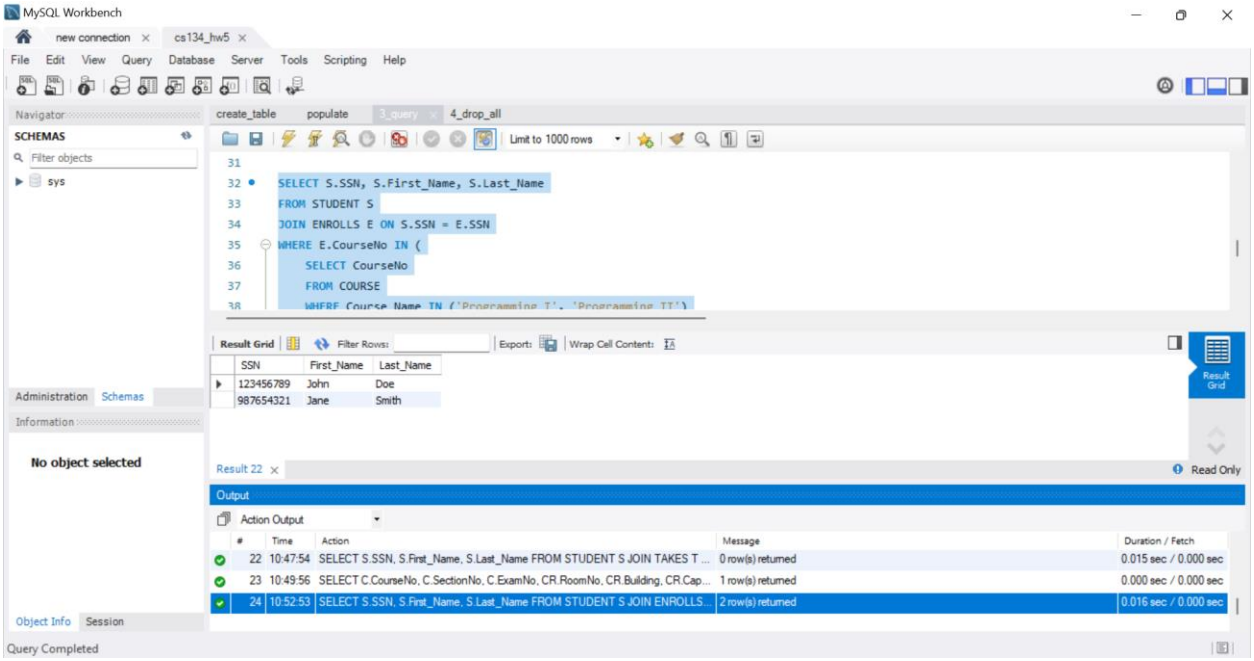
Query 5. Obtain information (exam number, course number and section number) about the exams taking place in building ‘Riverside Hall’. Along with that get the room’s capacity, room number and its building name.

```
SELECT C.CourseNo, C.SectionNo, C.ExamNo, CR.RoomNo, CR.Building, CR.Capacity
FROM CONDUCTED_IN C
JOIN CLASS_ROOM CR ON C.RoomNo = CR.RoomNo AND C.Building = CR.Building
WHERE CR.Building = 'Riverside Hall';
```



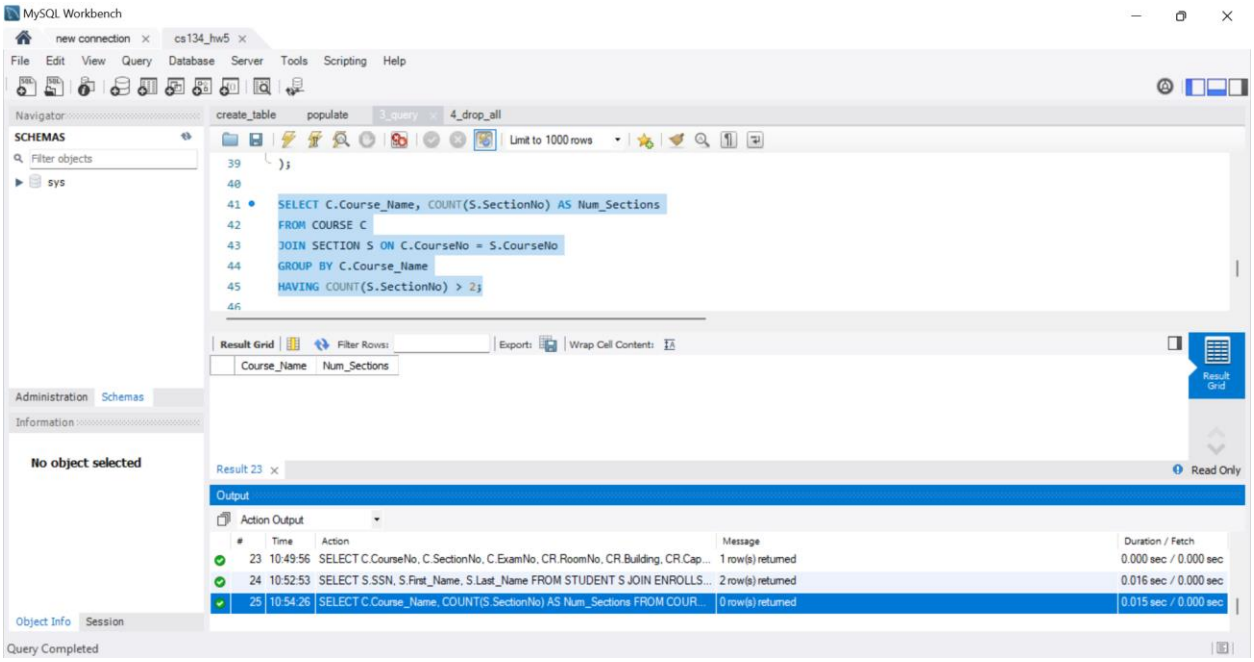
Query 6. Retrieve information (SSN, first name and last name) about students who are enrolled in the course of ‘Programming I’ or ‘Programming II’ (‘Programming I’ and ‘Programming II’ are course names).

```
SELECT S.SSN, S.First_Name, S.Last_Name
FROM STUDENT S
JOIN ENROLLS E ON S.SSN = E.SSN
WHERE E.CourseNo IN (
  SELECT CourseNo
  FROM COURSE
  WHERE Course_Name IN ('Programming I', 'Programming II')
);
```



Query 7. Retrieve the courses that have more than 2 sections. (List course name and number of sections)

```
SELECT C.Course_Name, COUNT(S.SectionNo) AS Num_Sections
FROM COURSE C
JOIN SECTION S ON C.CourseNo = S.CourseNo
GROUP BY C.Course_Name
HAVING COUNT(S.SectionNo) > 2;
```



Query 8. Retrieve all the exams that are conducted in RVR. “RVR” is build name. Retrieve the following information: Building, RoomNo, Capacity, CourseNo, SectionNo, ExamNo. If a classroom has no exam at all, display CourseNo, SectionNo, ExamNo with a NULL value. In your output, please show at least one classroom in RVR that has an exam and another classroom in RVR that has no exam.

```
SELECT CR.Building, CR.RoomNo, CR.Capacity, C.CourseNo, C.SectionNo, C.ExamNo
FROM CLASS_ROOM CR
LEFT JOIN CONDUCTED_IN C ON CR.RoomNo = C.RoomNo AND CR.Building = C.Building
WHERE CR.Building = 'RVR';
```

